

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460655.8523 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.13 +/- 0.01
Transit depth (Rp/Rs)^2	1.69 +/- 0.27 [%]
Orbital Inclination (inc)	80.3 +/- 2.04
Airmass coefficient 1 (a1)	0.9127 +/- 0.0058
Airmass coefficient 2 (a2)	0.086 +/- 0.033
Scatter in the residuals of the lightcurve fit is	0.63 %
Best Comparison Star	#1 - [83, 39]
Optimal Aperture	3.86
Optimal Annulus	10.81
Transit Duration (day)	0.1027 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.13 $\pm$ 0.01 vs 0.1178 (deviation 1.09 σ)
Duration fit vs published	0.1027 d vs 0.125 d (deviation 5.84 σ)
Mid-time O–C	2.0 min
Depth SNR	11.6
Residual scatter	0.63 %

Light curve

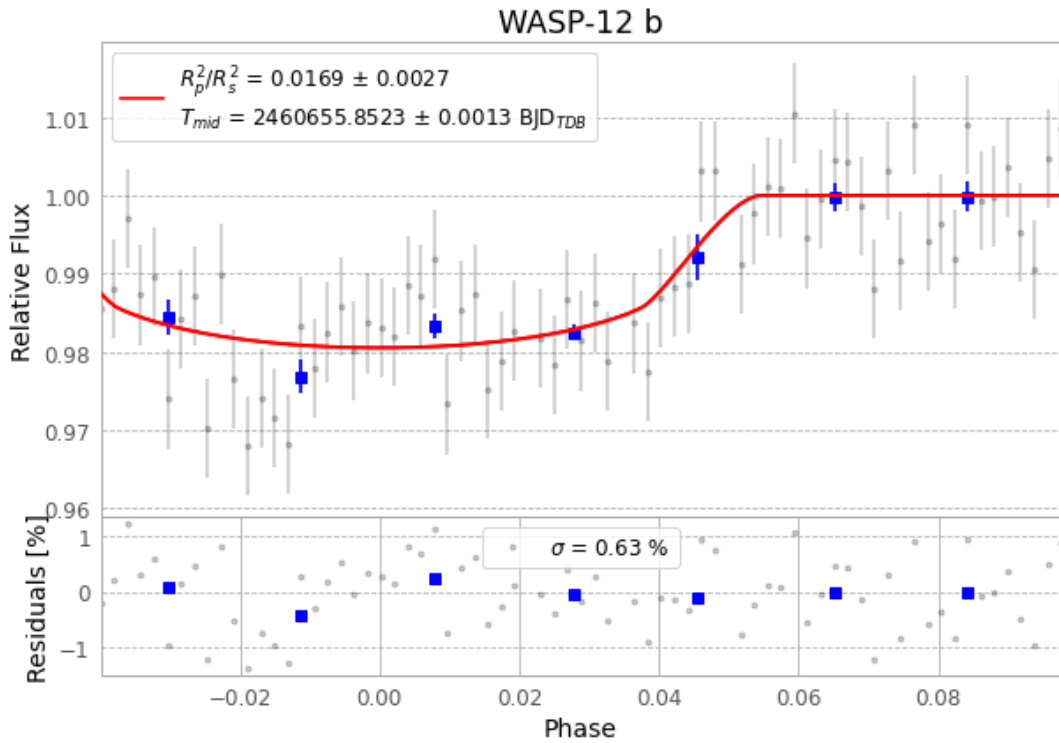


Figure 1 — FinalLightCurve_WASP-12 b_2024-12-11.png (SHA-256 dd9039b46cff049e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [420, 242], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ae9e0ad1eac5e33e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

73 FITS frames (48 MB), WASP-12241211071516.FITS → WASP-12241211105115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-241211.log (3dbefbd4bc4e...), FinalLightCurve_WASP-12 b_2024-12-11.png (dd9039b46cff...), FinalLightCurve_WASP-12 b_2024-12-11.csv (077676605e5b...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 04:11Z.